

ENUNCIADO — PROBLEMA 1

Numerical Analysis – Preliminary Exam

Department of Mathematical Sciences

George Mason University

AUGUST 2014

Instructions: Answer any **four** of the following **seven** problems. **Clearly indicate which four are to be graded.** This is a closed book, closed notes exam. Show all of your work for full credit. No calculators or computers are allowed.

- Let x_i^* , $i \geq 0$ denote positive numbers on a computer. With a unit round-off error δ , $x_i^* = fl(x_i) = x_i(1 + \epsilon_i)$ with $|\epsilon_i| \leq \delta$, where x_i are positive exact numbers.
 - Consider the product $P_n = \prod_{i=0}^n x_i$. Let the floating point approximation $fl(P_n) = P_n^* = P_n(1 + \epsilon)$. Prove that ϵ is bounded as $\epsilon \leq e^{\delta(2n+1)} - 1$.
 - Consider the expression $S = x_1 \cdot x_2 \cdot x_3 + x_4 \cdot x_5 \cdot x_6$. Let S^* be the floating point approximation of S . Prove that $\frac{S^*}{S} \leq e^{6\delta}$.
- Work out by hand, showing all necessary details and steps, the first Newton iterate x_1 for solving the problem $f(x) = 0$ where $f(x) = 1 - x - \ln x$ given an initial guess $x_0 = 1/2$.
 - Describe the convergence properties of Newton's method (to the exact root $x = 0$) applied to the function $f(x) = x^m$ where m is a positive integer. Be as specific as possible in terms of computing convergence rates.
 - Consider the nonlinear system $\vec{f}(\vec{x}) = 0$ where $\vec{f} = (f_1, f_2)$, $\vec{x} = (x, y)$ and

$$\begin{aligned} f_1(x, y) &= y - x^2 - c, \\ f_2(x, y) &= x^2 + y^2 - 1, \end{aligned}$$

where c is a constant to be specified.

For the case $c = 0$ write out by hand the specific linear system that must be solved in order to obtain the first Newton iterate $\vec{x}_1$ given the starting vector $\vec{x}_0 = (1/2, 1/2)$. Discuss convergence rates for this method and describe how these rates could be influenced by the value of the parameter c .

- Suppose that a function g maps the interval $[a, b]$ into itself and g satisfies a Lipschitz condition with a Lipschitz constant $0 \leq \lambda < 1$, then
 - Show that the sequence $x_{n+1} = g(x_n)$, $n = 0, 1, \dots$ (for any initial approximation $x_0 \in [a, b]$) converges to a unique fixed point ξ in $[a, b]$.
 - Moreover, prove the following error bound involved in using the sequence x_n to approximate ξ given by: $|x_n - \xi| \leq \frac{\lambda^n}{1 - \lambda} |x_1 - x_0| \quad \forall n \geq 1$.

SOLUCIÓN — PROBLEMA 1

Problema 1 — Análisis de error de redondeo en productos y sumas

Modelo estándar: para números exactos positivos x_i , su representación en punto flotante es $x_i^* = fl(x_i) = x_i(1 + \epsilon_i)$ con $|\epsilon_i| \leq \delta$ (δ = unidad de redondeo), y cada operación aritmética elemental introduce un factor adicional $(1 + \eta)$, $|\eta| \leq \delta$.

(a) Cota para el producto $P_n = \prod_{i=0}^n x_i$.

Calcular $fl(P_n)$ requiere: (i) almacenar los $n + 1$ factores, lo que introduce $n + 1$ factores de redondeo $(1 + \epsilon_i)$, $i = 0, \dots, n$; y (ii) realizar n multiplicaciones sucesivas para combinar $n + 1$ números, cada una introduciendo un factor $(1 + \eta_k)$, $k = 1, \dots, n$, $|\eta_k| \leq \delta$. En total,

$$P_n^* = fl(P_n) = P_n \prod_{i=0}^n (1 + \epsilon_i) \prod_{k=1}^n (1 + \eta_k) = P_n (1 + \epsilon),$$

$$1 + \epsilon = \prod_{i=0}^n (1 + \epsilon_i) \prod_{k=1}^n (1 + \eta_k),$$

un producto de exactamente $(n + 1) + n = 2n + 1$ factores, cada uno positivo (para $\delta < 1$) y acotado por $1 + \delta$.

Usando la desigualdad elemental $1 + t \leq e^t$ para todo $t \in \mathbb{R}$ (que se sigue de la convexidad de e^t , o de $e^t = 1 + t + t^2/2! + \dots \geq 1 + t$ para $t \geq 0$, y por continuidad para todo t), con $t = \delta$: $1 + \delta \leq e^\delta$. Por lo tanto, cada uno de los $2n + 1$ factores está acotado por e^δ , y

$$1 + \epsilon = \prod_{i=0}^n (1 + \epsilon_i) \prod_{k=1}^n (1 + \eta_k) \leq \underbrace{e^\delta \cdot e^\delta \cdots e^\delta}_{2n+1 \text{ factores}} = e^{(2n+1)\delta}.$$

Luego

$$\boxed{\epsilon \leq e^{\delta(2n+1)} - 1.} \quad \blacksquare$$

(b) Cota para $S = x_1 x_2 x_3 + x_4 x_5 x_6$.

Sea $A = x_1 x_2 x_3$ y $B = x_4 x_5 x_6$ (ambos productos de $m = 3$ factores). Por el resultado del inciso (a) aplicado con $m = 3$ números (es decir, $n+1=3 \rightarrow n=2$, dando $2n+1=5$ fuentes de redondeo: 3 de almacenamiento + 2 de multiplicación):

$$A^* = fl(A) = A(1 + \alpha), \quad 1 + \alpha \leq e^{5\delta},$$

$$B^* = fl(B) = B(1 + \beta), \quad 1 + \beta \leq e^{5\delta}.$$

La suma final introduce un redondeo adicional $(1 + \gamma)$, $|\gamma| \leq \delta$, con $1 + \gamma \leq e^\delta$:

$$S^* = fl(A^* + B^*) = (A^* + B^*)(1 + \gamma).$$

Como $A, B > 0$ (los x_i son positivos) y $1 + \alpha, 1 + \beta \leq e^{5\delta}$:

$$A^* + B^* = A(1 + \alpha) + B(1 + \beta) \leq e^{5\delta}(A + B) = e^{5\delta}S.$$

Por lo tanto

$$S^* = (A^* + B^*)(1 + \gamma) \leq e^{5\delta}S \cdot e^\delta = e^{6\delta}S,$$

es decir,

$$\boxed{\frac{S^*}{S} \leq e^{6\delta}}. \quad \blacksquare$$

Verificación: el exponente $6 = 5 + 1$ tiene sentido: 5 proviene de las fuentes de error de **cada** triple producto (que comparten el mismo factor de cota $e^{5\delta}$, por lo que no se suman entre sí al acotar la suma $A + B$, sino que factorizan), y el +1 final proviene de la única suma adicional que combina ambos productos.

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SOLUCIÓN — PROBLEMA 2

Problema 2 — Newton y raíces múltiples

(a) Primera iterada de Newton para $f(x) = 1 - x - \ln x$, $x_0 = 1/2$.

$f'(x) = -1 - \frac{1}{x}$. Entonces

$$f(1/2) = 1 - \frac{1}{2} - \ln(1/2) = \frac{1}{2} + \ln 2, \quad f'(1/2) = -1 - 2 = -3.$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{1}{2} - \frac{\frac{1}{2} + \ln 2}{-3} = \frac{1}{2} + \frac{\frac{1}{2} + \ln 2}{3} = \frac{1}{2} + \frac{1}{6} + \frac{\ln 2}{3} = \frac{2}{3} + \frac{\ln 2}{3}.$$

$$x_1 = \frac{2 + \ln 2}{3} \approx \frac{2 + 0.693147}{3} \approx 0.897716.$$

Verificación: $f(x_1) = 1 - 0.897716 - \ln(0.897716) \approx 0.102284 - (-0.107942) \approx 0.0102$, *muchomáspequeñoque* $f(x_0) \approx 1.193$: consistente con un paso de Newton acercándose a la raíz (la raíz exacta de $1 - x - \ln x = 0$ es $x^* = 1$, y en efecto $x_1 \approx 0.898$ está más cerca de 1 que $x_0 = 0.5$).

(b) Convergencia de Newton para $f(x) = x^m$ (raíz $x = 0$ de multiplicidad m).

$f'(x) = mx^{m-1}$. La iteración de Newton es

$$x_{n+1} = x_n - \frac{x_n^m}{mx_n^{m-1}} = x_n - \frac{x_n}{m} = x_n \left(1 - \frac{1}{m}\right) = x_n \cdot \frac{m-1}{m}.$$

Por inducción, $x_n = x_0 \left(\frac{m-1}{m}\right)^n \rightarrow 0$ cuando $n \rightarrow \infty$ (pues $0 \leq \frac{m-1}{m} < 1$ para $m \geq 1$ entero).

Con $e_n := x_n - 0 = x_n$:

$$e_{n+1} = \frac{m-1}{m} e_n,$$

es decir, la **convergencia es lineal** (orden 1), con constante asintótica de error

$$C = \frac{m-1}{m} = 1 - \frac{1}{m} \in [0, 1),$$

en contraste con la convergencia **cuadrática** que Newton exhibe en raíces **simples** ($m = 1$; en ese caso, de hecho, $C = 0$ y $x_1 = 0$ exactamente, pues la ecuación ya es lineal). Cuanto mayor es la multiplicidad m , más cerca de 1 está C y más lenta es la convergencia lineal.

(c) Sistema no lineal 2×2 con $c = 0$.

$f_1(x, y) = y - x^2$, $f_2(x, y) = x^2 + y^2 - 1$. El Jacobiano es

$$J(x, y) = \begin{pmatrix} \partial f_1 / \partial x & \partial f_1 / \partial y \\ \partial f_2 / \partial x & \partial f_2 / \partial y \end{pmatrix} = \begin{pmatrix} -2x & 1 \\ 2x & 2y \end{pmatrix}.$$

En $\vec{x}_0 = (1/2, 1/2)$: $J_0 = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$, y

$$f_1(1/2, 1/2) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}, \quad f_2(1/2, 1/2) = \frac{1}{4} + \frac{1}{4} - 1 = -\frac{1}{2}.$$

El sistema de Newton $J_0 \Delta = -\vec{f}(\vec{x}_0)$ es

$$\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} = \begin{pmatrix} -1/4 \\ 1/2 \end{pmatrix}.$$

Sumando ambas ecuaciones ($-\Delta x + \Delta y = -\frac{1}{4}$ y $\Delta x + \Delta y = \frac{1}{2}$): $2\Delta y = \frac{1}{4} \Rightarrow \Delta y = \frac{1}{8}$, y $\Delta x = \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$. Por lo tanto

$$\vec{x}_1 = \vec{x}_0 + \Delta = \left(\frac{1}{2} + \frac{3}{8}, \frac{1}{2} + \frac{1}{8} \right) = \boxed{\left(\frac{7}{8}, \frac{5}{8} \right) = (0.875, 0.625)}.$$

Verificación: sustituyendo en las ecuaciones originales del sistema lineal, $-\frac{3}{8} + \frac{1}{8} = -\frac{1}{4}$ ✓, $\frac{3}{8} + \frac{1}{8} = \frac{1}{2}$ ✓.

Discusión de convergencia. La solución exacta del sistema ($y = x^2$, $x^2 + y^2 = 1 \Rightarrow x^4 + x^2 - 1 = 0$) da $x^2 = \frac{-1+\sqrt{5}}{2} \approx 0.618$, $x \approx 0.786$, $y \approx 0.618$. Como $f_1, f_2 \in C^\infty$ y $\det J = -4xy - 2x = -2x(2y + 1) \neq 0$ en esa solución (pues $x \approx 0.786 \neq 0$ y $2y + 1 \approx 2.236 \neq 0$), el teorema de Newton–Kantorovich garantiza convergencia **cuadrática local** (para $\vec{x}_0$ suficientemente cerca de la solución).

El parámetro c desplaza verticalmente la parábola $y = x^2 + c$ respecto al círculo unitario, cambiando la posición y el ángulo de intersección de las dos curvas. La velocidad de convergencia de Newton depende de que J permanezca no singular cerca de la raíz: $\det J = -2x(2y + 1)$ se anula cuando $x = 0$ o $y = -1/2$, situaciones que corresponden geométricamente a que la parábola sea **tangente** al círculo en vez de cortarlo transversalmente. Para valores de c cercanos a esos valores críticos (donde las dos curvas casi se tocan tangencialmente), el Jacobiano en la raíz se vuelve casi singular, y la convergencia cuadrática de Newton se degrada (se vuelve lenta o incluso puede fallar), mientras que para valores de c que dan intersecciones bien transversales, se preserva la convergencia cuadrática.

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 - (b) Consider the expression $S = x_1 \cdot x_2 \cdot x_3 + x_4 \cdot x_5 \cdot x_6$. Let S^* be the floating point approximation of S . Prove that $\frac{S^*}{S} \leq e^{6\delta}$.
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 - (b) Describe the convergence properties of Newton's method (to the exact root $x = 0$) applied to the function $f(x) = x^m$ where m is a positive integer. Be as specific as possible in terms of computing convergence rates.
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3. Suppose that a function g maps the interval $[a, b]$ into itself and g satisfies a Lipschitz condition with a Lipschitz constant $0 \leq \lambda < 1$, then
 - (a) Show that the sequence $x_{n+1} = g(x_n)$, $n = 0, 1, \dots$ (for any initial approximation $x_0 \in [a, b]$) converges to a unique fixed point ξ in $[a, b]$.
 - (b) Moreover, prove the following error bound involved in using the sequence x_n to approximate ξ given by: $|x_n - \xi| \leq \frac{\lambda^n}{1 - \lambda} |x_1 - x_0| \quad \forall n \geq 1$.

4. (a) Derive the two point Gaussian integration formula $G_2(f)$ for $I(f) = \int_{-1}^1 f(x)dx$.
 (b) Indicate for what polynomial degree this rule is exact. Use the two-point Gauss Quadrature formula derived to evaluate the integral $\int_1^4 \frac{dx}{5x+1}$.
5. (a) Let f'' be continuous in $[a, b]$ and let $a = t_0 < t_1 < \dots < t_n = b$. If S is a complete cubic spline interpolating f at the knots t_i for $0 \leq i \leq n$, then prove that,

$$\int_a^b [S''(x)]^2 dx \leq \int_a^b [f''(x)]^2 dx$$

- (b) A complete cubic spline S for a function f is defined on $[1, 3]$ by,

$$s(x) = \begin{cases} s_0(x) = 3(x-1) + 2(x-1)^2 - (x-1)^3, & \text{if } 1 \leq x < 2 \\ s_1(x) = a + b(x-2) + c(x-2)^2 + d(x-2)^3 & \text{if } 2 \leq x \leq 3 \end{cases}$$

Given $f'(1) = f'(3)$, find a, b, c, d .

6. Determine the total number of operations (addition, subtraction, multiplication, division) for the Gaussian Elimination algorithm described below:

```

input  $n, (a_{ij})$ 
for  $k = 1$  to  $n - 1$  do
  for  $i = k + 1$  to  $n$  do
     $z = \frac{a_{ik}}{a_{kk}}$ 
    for  $j = k$  to  $n$  do
       $a_{ij} = a_{ij} - za_{kj}$ 
    end do
  end do
end do
```

7. Consider the initial-value problem $y'(t) = f(t, y)$ for $0 \leq t \leq 1$ with $y(0) = a$. Consider the one-step method

$$y_{k+1} = y_k + h \Phi(t_k, y_k, h)$$

with $y_0 = a$, $h = 1/N$, and $t_k = kh$ for $k = 0, 1, \dots, N$. Assume that there is a constant L such that $|\Phi(t, y, h) - \Phi(t, z, h)| \leq L |y - z|$ for all $t, y, z \in \mathcal{R}$. Furthermore, assume that the solution $y(t)$ satisfies $|y(t+h) - y(t) - h\Phi(t, y(t), h)| \leq c h^{p+1}$ for all $t, h \in [0, 1]$. Prove that

$$|y_N - y(1)| \leq c \frac{h^p}{L} (e^L - 1).$$

SOLUCIÓN — PROBLEMA 3

Problema 3 — Teorema del punto fijo (contracción)

Sea $g : [a, b] \rightarrow [a, b]$ con constante de Lipschitz $0 \leq \lambda < 1$: $|g(x) - g(y)| \leq \lambda|x - y|$ para todo $x, y \in [a, b]$.

(a) Existencia y unicidad del punto fijo.

Sea $x_0 \in [a, b]$ arbitrario y $x_{n+1} = g(x_n)$. Como g mapea $[a, b]$ en sí mismo, $x_n \in [a, b]$ para todo n .

Cota de pasos sucesivos. Por la condición de Lipschitz,

$$|x_{n+1} - x_n| = |g(x_n) - g(x_{n-1})| \leq \lambda|x_n - x_{n-1}| \leq \dots \leq \lambda^n|x_1 - x_0|.$$

La sucesión es de Cauchy. Para $m > n$, por la desigualdad triangular y la suma geométrica:

$$|x_m - x_n| \leq \sum_{k=n}^{m-1} |x_{k+1} - x_k| \leq \sum_{k=n}^{m-1} \lambda^k |x_1 - x_0| = \lambda^n |x_1 - x_0| \sum_{j=0}^{m-n-1} \lambda^j \leq \frac{\lambda^n}{1 - \lambda} |x_1 - x_0|.$$

Como $0 \leq \lambda < 1$, $\lambda^n \rightarrow 0$, y el lado derecho $\rightarrow 0$ cuando $n \rightarrow \infty$: $\{x_n\}$ es de Cauchy en $\mathbb{R}$ (completo), luego converge a algún ξ . Como $[a, b]$ es cerrado, $\xi \in [a, b]$.

ξ es punto fijo. Como g es Lipschitz, es continua; tomando límite en $x_{n+1} = g(x_n)$: $\xi = g(\xi)$.

Unicidad. Si $\xi, \eta \in [a, b]$ son ambos puntos fijos con $\xi \neq \eta$, entonces $|\xi - \eta| = |g(\xi) - g(\eta)| \leq \lambda|\xi - \eta| < |\xi - \eta|$ (pues $\lambda < 1$ y $|\xi - \eta| > 0$), contradicción. Luego $\xi = \eta$. ■

(b) Cota de error a priori.

De la desigualdad de Cauchy anterior, tomando el límite $m \rightarrow \infty$ (la norma es continua):

$$|\xi - x_n| = \lim_{m \rightarrow \infty} |x_m - x_n| \leq \frac{\lambda^n}{1 - \lambda} |x_1 - x_0|.$$

Es decir,

$$|x_n - \xi| \leq \frac{\lambda^n}{1 - \lambda} |x_1 - x_0| \quad \forall n \geq 1. \quad \blacksquare$$

ENUNCIADO — PROBLEMA 4

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$$|y_N - y(1)| \leq c \frac{h^p}{L} (e^L - 1).$$

SOLUCIÓN — PROBLEMA 4

Problema 4 — Cuadratura gaussiana de dos puntos (evaluación)

(a)-(b) Derivación de G_2 y grado de exactitud.(Idéntico al Problema 2(a) del examen de Enero 2014.) Por simetría, se buscan $x_1 = -x_2 = t$, $w_1 = w_2 = w$, exactos para $1, x, x^2, x^3$:

- $f = 1$: $2w = 2 \Rightarrow w = 1$.
- $f = x$: automático por simetría.
- $f = x^2$: $2wt^2 = \frac{2}{3} \Rightarrow t = \frac{1}{\sqrt{3}}$.
- $f = x^3$: automático por simetría (integrando impar).

$$G_2(f) = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right),$$

exacta para **polinomios de grado** ≤ 3 (grado máximo $2n - 1 = 3$ para $n = 2$ nodos, alcanzado porque $\pm 1/\sqrt{3}$ son las raíces del polinomio de Legendre $P_2(x) = \frac{1}{2}(3x^2 - 1)$).

Evaluación de $\int_1^4 \frac{dx}{5x+1}$.

Cambio de variable $x = \frac{3}{2}t + \frac{5}{2}$, $dx = \frac{3}{2}dt$:

$$5x+1 = 5\left(\frac{3}{2}t + \frac{5}{2}\right) + 1 = \frac{15}{2}t + \frac{27}{2} = \frac{15t+27}{2}.$$

$$\int_1^4 \frac{dx}{5x+1} = \int_{-1}^1 \frac{\frac{3}{2}dt}{\frac{15t+27}{2}} = \int_{-1}^1 \frac{3dt}{15t+27} = \int_{-1}^1 \frac{dt}{5t+9}.$$

Aplicando G_2 con $t = \pm 1/\sqrt{3}$:

$$G_2 = \frac{1}{9 + \frac{5}{\sqrt{3}}} + \frac{1}{9 - \frac{5}{\sqrt{3}}} = \frac{\left(9 - \frac{5}{\sqrt{3}}\right) + \left(9 + \frac{5}{\sqrt{3}}\right)}{81 - \frac{25}{3}} = \frac{18}{\frac{243-25}{3}} = \frac{18 \cdot 3}{218} = \frac{54}{218} = \boxed{\frac{27}{109} \approx 0.24771}.$$

Verificación: el valor exacto es

$$\int_1^4 \frac{dx}{5x+1} = \frac{1}{5} \ln(5x+1) \Big|_1^4 = \frac{1}{5} (\ln 21 - \ln 6) = \frac{1}{5} \ln\left(\frac{7}{2}\right) \approx \frac{1.252763}{5} \approx 0.250553.$$

La diferencia $|0.24771 - 0.250553| \approx 0.0028$ es pequeña, consistente con el error de una regla de Gauss de 2 puntos aplicada a una función suave (aunque de mayor curvatura que en el Problema 2 de enero, dado el factor 5).

ENUNCIADO — PROBLEMA 5

4. (a) Derive the two point Gaussian integration formula $G_2(f)$ for $I(f) = \int_{-1}^1 f(x)dx$.
 (b) Indicate for what polynomial degree this rule is exact. Use the the two-point Gauss Quadrature formula derived to evaluate the integral $\int_1^4 \frac{dx}{5x+1}$.
5. (a) Let f'' be continuous in $[a, b]$ and let $a = t_0 < t_1 < \dots < t_n = b$. If S is a complete cubic spline interpolating f at the knots t_i for $0 \leq i \leq n$, then prove that,

$$\int_a^b [S''(x)]^2 dx \leq \int_a^b [f''(x)]^2 dx$$

- (b) A complete cubic spline S for a function f is defined on $[1, 3]$ by,

$$s(x) = \begin{cases} s_0(x) = 3(x-1) + 2(x-1)^2 - (x-1)^3, & \text{if } 1 \leq x < 2 \\ s_1(x) = a + b(x-2) + c(x-2)^2 + d(x-2)^3 & \text{if } 2 \leq x \leq 3 \end{cases}$$

Given $f'(1) = f'(3)$, find a, b, c, d .

6. Determine the total number of operations (addition, subtraction, multiplication, division) for the Gaussian Elimination algorithm described below:

```

input  $n, (a_{ij})$ 
for  $k = 1$  to  $n - 1$  do
  for  $i = k + 1$  to  $n$  do
     $z = \frac{a_{ik}}{a_{kk}}$ 
    for  $j = k$  to  $n$  do
       $a_{ij} = a_{ij} - za_{kj}$ 
    end do
  end do
end do
```

7. Consider the initial-value problem $y'(t) = f(t, y)$ for $0 \leq t \leq 1$ with $y(0) = a$. Consider the one-step method

$$y_{k+1} = y_k + h \Phi(t_k, y_k, h)$$

with $y_0 = a$, $h = 1/N$, and $t_k = kh$ for $k = 0, 1, \dots, N$. Assume that there is a constant L such that $|\Phi(t, y, h) - \Phi(t, z, h)| \leq L |y - z|$ for all $t, y, z \in \mathcal{R}$. Furthermore, assume that the solution $y(t)$ satisfies $|y(t+h) - y(t) - h\Phi(t, y(t), h)| \leq c h^{p+1}$ for all $t, h \in [0, 1]$. Prove that

$$|y_N - y(1)| \leq c \frac{h^p}{L} (e^L - 1).$$

SOLUCIÓN — PROBLEMA 5

Problema 5 — Splines cúbicos completos

(a) Propiedad de mínima curvatura del spline cúbico completo.

Sea S el spline cúbico completo que interpola $f \in C^2[a, b]$ en los nudos $t_0 < \dots < t_n$, con $S(t_i) = f(t_i)$ y **condiciones de frontera de spline completo** $S'(a) = f'(a)$, $S'(b) = f'(b)$. Sea $e = f - S$.

Expandiendo el cuadrado:

$$\int_a^b (f'')^2 dx = \int_a^b (S'' + e'')^2 dx = \int_a^b (S'')^2 dx + 2 \int_a^b S'' e'' dx + \int_a^b (e'')^2 dx. \quad (\dagger)$$

Afirmación: $\int_a^b S'' e'' dx = 0$.

En cada subintervalo (t_{i-1}, t_i) , S es cúbica, así que S''' es **constante** allí (llamémosla c_i). Integrando por partes:

$$\int_{t_{i-1}}^{t_i} S'' e'' dx = [S'' e']_{t_{i-1}}^{t_i} - \int_{t_{i-1}}^{t_i} S''' e' dx = [S'' e']_{t_{i-1}}^{t_i} - c_i \int_{t_{i-1}}^{t_i} e' dx.$$

Como $e(t_i) = f(t_i) - S(t_i) = 0$ en **todos** los nudos (condición de interpolación),

$$\int_{t_{i-1}}^{t_i} e' dx = e(t_i) - e(t_{i-1}) = 0.$$

Luego $\int_{t_{i-1}}^{t_i} S'' e'' dx = S''(t_i) e'(t_i) - S''(t_{i-1}) e'(t_{i-1})$. Sumando sobre $i = 1, \dots, n$ (la suma telescopa, ya que S'' y $e' = f' - S'$ son continuas en $[a, b]$ porque $S \in C^2$ y $f \in C^2$):

$$\int_a^b S'' e'' dx = S''(t_n) e'(t_n) - S''(t_0) e'(t_0) = S''(b) e'(b) - S''(a) e'(a).$$

Por las condiciones de spline completo, $e'(a) = f'(a) - S'(a) = 0$ y $e'(b) = f'(b) - S'(b) = 0$.

Ambos términos de frontera se anulan, así que

$$\int_a^b S'' e'' dx = 0.$$

Sustituyendo en $(\dagger)$:

$$\int_a^b (f'')^2 dx = \int_a^b (S'')^2 dx + \int_a^b (e'')^2 dx \geq \int_a^b (S'')^2 dx,$$

pues $\int_a^b (e'')^2 dx \geq 0$. ■

(b) Determinación de a, b, c, d .

$s_0(x) = 3(x-1) + 2(x-1)^2 - (x-1)^3$ en $[1, 2)$; $s_1(x) = a + b(x-2) + c(x-2)^2 + d(x-2)^3$ en $[2, 3]$. Se necesita continuidad C^2 en el nudo interior $x = 2$, más la condición de frontera de spline completo que, junto con el dato $f'(1) = f'(3)$, impone $S'(1) = S'(3)$.

Derivadas de s_0 :

$$s'_0(x) = 3 + 4(x-1) - 3(x-1)^2, \quad s''_0(x) = 4 - 6(x-1).$$

Evaluando en $x = 2$ (es decir, $x-1 = 1$):

$$s_0(2) = 3 + 2 - 1 = 4, \quad s'_0(2) = 3 + 4 - 3 = 4, \quad s''_0(2) = 4 - 6 = -2.$$

Continuidad C^0 en $x = 2$: $s_1(2) = a = s_0(2) = 4 \Rightarrow \boxed{a = 4}$.

Continuidad C^1 : $s'_1(x) = b + 2c(x-2) + 3d(x-2)^2$, así $s'_1(2) = b$. Igualando a $s'_0(2) = 4$: $\boxed{b = 4}$.

Continuidad C^2 : $s''_1(x) = 2c + 6d(x-2)$, así $s''_1(2) = 2c$. Igualando a $s''_0(2) = -2$: $2c = -2 \Rightarrow \boxed{c = -1}$.

Condición de frontera $S'(1) = S'(3)$ (equivalente a $f'(1) = f'(3)$ para el spline completo): $s'_0(1) = 3$ (evaluando en $x-1 = 0$). Y

$$s'_1(3) = b + 2c(1) + 3d(1)^2 = 4 - 2 + 3d = 2 + 3d.$$

Igualando: $3 = 2 + 3d \Rightarrow d = \frac{1}{3}$, es decir $\boxed{d = \frac{1}{3}}$.

Resumen: $a = 4$, $b = 4$, $c = -1$, $d = \frac{1}{3}$.

Verificación (conteo de grados de libertad): s_0 está completamente dado; s_1 tiene 4 incógnitas (a, b, c, d), y se impusieron exactamente 4 condiciones (continuidad de s, s', s'' en $x = 2$, más la condición de frontera): sistema determinado de forma única, consistente con la teoría de splines cúbicos completos (2 nudos interiores... aquí 1 nudo interior + 2 condiciones de frontera = $n + 1 = 2$ ecuaciones adicionales a las $4(n-1)$ de continuidad interna, exactamente lo necesario para determinar los $4n$ coeficientes con $n = 2$ tramos).

ENUNCIADO — PROBLEMA 6

4. (a) Derive the two point Gaussian integration formula $G_2(f)$ for $I(f) = \int_{-1}^1 f(x)dx$.
 (b) Indicate for what polynomial degree this rule is exact. Use the the two-point Gauss Quadrature formula derived to evaluate the integral $\int_1^4 \frac{dx}{5x+1}$.
5. (a) Let f'' be continuous in $[a, b]$ and let $a = t_0 < t_1 < \dots < t_n = b$. If S is a complete cubic spline interpolating f at the knots t_i for $0 \leq i \leq n$, then prove that,

$$\int_a^b [S''(x)]^2 dx \leq \int_a^b [f''(x)]^2 dx$$

- (b) A complete cubic spline S for a function f is defined on $[1, 3]$ by,

$$s(x) = \begin{cases} s_0(x) = 3(x-1) + 2(x-1)^2 - (x-1)^3, & \text{if } 1 \leq x < 2 \\ s_1(x) = a + b(x-2) + c(x-2)^2 + d(x-2)^3 & \text{if } 2 \leq x \leq 3 \end{cases}$$

Given $f'(1) = f'(3)$, find a, b, c, d .

6. Determine the total number of operations (addition, subtraction, multiplication, division) for the Gaussian Elimination algorithm described below:

```

input  $n, (a_{ij})$ 
for  $k = 1$  to  $n - 1$  do
    for  $i = k + 1$  to  $n$  do
         $z = \frac{a_{ik}}{a_{kk}}$ 
        for  $j = k$  to  $n$  do
             $a_{ij} = a_{ij} - za_{kj}$ 
        end do
    end do
end do
```

7. Consider the initial-value problem $y'(t) = f(t, y)$ for $0 \leq t \leq 1$ with $y(0) = a$. Consider the one-step method

$$y_{k+1} = y_k + h \Phi(t_k, y_k, h)$$

with $y_0 = a$, $h = 1/N$, and $t_k = kh$ for $k = 0, 1, \dots, N$. Assume that there is a constant L such that $|\Phi(t, y, h) - \Phi(t, z, h)| \leq L |y - z|$ for all $t, y, z \in \mathcal{R}$. Furthermore, assume that the solution $y(t)$ satisfies $|y(t+h) - y(t) - h\Phi(t, y(t), h)| \leq c h^{p+1}$ for all $t, h \in [0, 1]$. Prove that

$$|y_N - y(1)| \leq c \frac{h^p}{L} (e^L - 1).$$

SOLUCIÓN — PROBLEMA 6

Problema 6 — Conteo de operaciones en eliminación gaussiana

Algoritmo dado:

```
for k = 1 to n-1   for i = k+1 to n       z = a_ik / a_kk
```

Conteo para k fijo. El bucle en i recorre $i = k + 1, \dots, n$: $(n - k)$ valores. Para cada i : 1 división (calcular z), y el bucle en j recorre $j = k, \dots, n$: $(n - k + 1)$ valores, cada uno con 1 multiplicación y 1 resta.

Entonces, para k fijo:

$$\# \text{divisiones} = n - k, \quad \# \text{mult} = \# \text{restas} = (n - k)(n - k + 1).$$

Suma sobre $k = 1, \dots, n - 1$. Sea $m = n - k$, que recorre $m = 1, \dots, n - 1$ cuando $k = 1, \dots, n - 1$.

Divisiones:

$$\sum_{k=1}^{n-1} (n - k) = \sum_{m=1}^{n-1} m = \frac{(n - 1)n}{2}.$$

Multiplicaciones (= restas):

$$\sum_{m=1}^{n-1} m(m + 1) = \sum_{m=1}^{n-1} m^2 + \sum_{m=1}^{n-1} m = \frac{(n - 1)n(2n - 1)}{6} + \frac{(n - 1)n}{2} = \frac{(n - 1)n[(2n - 1) + 3]}{6} = \frac{(n - 1)n \cdot 2(n + 1)}{6} = \frac{n(n - 1)(n + 1)}{3}.$$

Resumen:

$$\# \text{divisiones} = \frac{n(n - 1)}{2}, \quad \# \text{multiplicaciones} = \# \text{restas} = \frac{n(n - 1)(n + 1)}{3} = \frac{n(n^2 - 1)}{3}.$$

Total de operaciones aritméticas (sumas/restas + multiplicaciones + divisiones):

$$T(n) = \frac{n(n - 1)}{2} + 2 \cdot \frac{n(n^2 - 1)}{3} = n(n - 1) \left[\frac{1}{2} + \frac{2(n + 1)}{3} \right] = n(n - 1) \cdot \frac{3 + 4(n + 1)}{6}$$

$$T(n) = \frac{n(n - 1)(4n + 7)}{6}.$$

Verificación con $n = 2$: el algoritmo tiene solo $k = 1$, $i = 2$: 1 división (para z), y $j = 1, 2$ dan 2 multiplicaciones y 2 restas. Total = 5. Fórmula: $T(2) = \frac{2 \cdot 1 \cdot 15}{6} = \frac{30}{6} = 5 \checkmark$. También, $\# \text{divisiones} = \frac{2 \cdot 1}{2} = 1 \checkmark$, $\# \text{mult} = \frac{2 \cdot 3}{3} = 2 \checkmark$.

Verificación asintótica: para n grande, $T(n) \approx \frac{4n^3}{6} = \frac{2n^3}{3}$, coincidiendo con el conteo clásico de $\frac{2n^3}{3} + O(n^2)$ operaciones de punto flotante para la fase de eliminación (triangularización) de Gauss sin sustitución hacia atrás, el resultado estándar de la literatura.

ENUNCIADO — PROBLEMA 7

4. (a) Derive the two point Gaussian integration formula $G_2(f)$ for $I(f) = \int_{-1}^1 f(x)dx$.
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        for  $j = k$  to  $n$  do
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    end do
end do

```

7. Consider the initial-value problem $y'(t) = f(t, y)$ for $0 \leq t \leq 1$ with $y(0) = a$. Consider the one-step method

$$y_{k+1} = y_k + h \Phi(t_k, y_k, h)$$

with $y_0 = a$, $h = 1/N$, and $t_k = kh$ for $k = 0, 1, \dots, N$. Assume that there is a constant L such that $|\Phi(t, y, h) - \Phi(t, z, h)| \leq L |y - z|$ for all $t, y, z \in \mathcal{R}$. Furthermore, assume that the solution $y(t)$ satisfies $|y(t+h) - y(t) - h\Phi(t, y(t), h)| \leq c h^{p+1}$ for all $t, h \in [0, 1]$. Prove that

$$|y_N - y(1)| \leq c \frac{h^p}{L} (e^L - 1).$$

SOLUCIÓN — PROBLEMA 7

Problema 7 — Convergencia de métodos de un paso para PVI

Método de un paso: $y_{k+1} = y_k + h \Phi(t_k, y_k, h)$, $y_0 = a = y(0)$, $h = 1/N$, $t_k = kh$. Hipótesis: Φ Lipschitz en y con constante L ($|\Phi(t, y, h) - \Phi(t, z, h)| \leq L|y - z|$), y error de truncamiento local acotado:

$$\tau_k := y(t_{k+1}) - y(t_k) - h \Phi(t_k, y(t_k), h), \quad |\tau_k| \leq ch^{p+1}.$$

Objetivo: probar $|y_N - y(1)| \leq \frac{ch^p}{L}(e^L - 1)$.

Recurrencia del error global. Sea $e_k := y_k - y(t_k)$ (con $e_0 = 0$, pues $y_0 = a = y(0)$). Restando las definiciones de y_{k+1} e $y(t_{k+1})$:

$$\begin{aligned} e_{k+1} &= y_{k+1} - y(t_{k+1}) = [y_k + h\Phi(t_k, y_k, h)] - [y(t_k) + h\Phi(t_k, y(t_k), h) + \tau_k] \\ &= e_k + h[\Phi(t_k, y_k, h) - \Phi(t_k, y(t_k), h)] - \tau_k. \end{aligned}$$

Tomando valor absoluto, usando la condición de Lipschitz y la desigualdad triangular:

$$|e_{k+1}| \leq |e_k| + hL|e_k| + |\tau_k| \leq (1 + hL)|e_k| + ch^{p+1}. \quad (\star)$$

Cota por inducción. Afirmamos que

$$|e_k| \leq \frac{ch^p}{L} [(1 + hL)^k - 1] \quad \forall k \geq 0. \quad (\ddagger)$$

Base $k = 0$: el lado derecho es $\frac{ch^p}{L}[1 - 1] = 0 = |e_0|$ ✓.

Paso inductivo: suponiendo $(\ddagger)$ para k , usando $(\star)$:

$$\begin{aligned} |e_{k+1}| &\leq (1 + hL)|e_k| + ch^{p+1} \leq (1 + hL) \cdot \frac{ch^p}{L} [(1 + hL)^k - 1] + ch^{p+1} \\ &= \frac{ch^p}{L} (1 + hL)^{k+1} - \frac{ch^p}{L} (1 + hL) + ch^{p+1} = \frac{ch^p}{L} (1 + hL)^{k+1} - \frac{ch^p}{L} - ch^{p+1} + ch^{p+1} \\ &= \frac{ch^p}{L} [(1 + hL)^{k+1} - 1], \end{aligned}$$

que es $(\ddagger)$ para $k + 1$. Por inducción, $(\ddagger)$ vale para todo $k \geq 0$. ■ (inducción completa)

Evaluación en $k = N$. Como $h = 1/N$, se tiene $Nh = 1$. Usando de nuevo $1 + t \leq e^t$ con $t = hL$:

$$(1 + hL)^N \leq e^{NhL} = e^L.$$

Sustituyendo en $(\ddagger)$ con $k = N$ (y $t_N = Nh = 1$, así que $y(t_N) = y(1)$):

$$|y_N - y(1)| = |e_N| \leq \frac{ch^p}{L} [(1 + hL)^N - 1] \leq \frac{ch^p}{L} (e^L - 1).$$

$$|y_N - y(1)| \leq \frac{c h^p}{L} (e^L - 1). \quad \blacksquare$$

Verificación (caso límite): si $\tau_k = 0$ para todo k (es decir, $c = 0$, el método reproduce la solución exacta en cada paso), la cota da $|e_N| \leq 0$, lo cual es correcto ($e_k \equiv 0$ por inducción trivial en $(\star)$). Además, la cota crece con L (problemas más "rígidos"/sensibles amplifican más el error) y mejora (disminuye) con h^p conforme $h \rightarrow 0$, consistente con que un método de orden p más alto y un paso más fino dan mejor convergencia — este es precisamente el resultado clásico de Lipschitz + consistencia $\Rightarrow$ convergencia (teorema de Lax-tipo para PVI de un paso).